

Yucheng (Cecilia) Zhang

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EDUCATION

Northwestern University

M.S. in Computer Science

Aug. 2026 – May 2028

Evanston, IL

The University of Texas at Austin

B.S. in Mathematics, Minor in Computer Science | GPA: 3.74/4.00

Aug. 2023 – May 2026

Austin, TX

TECHNICAL SKILLS

ML/AI: Scikit-learn, PyTorch, TensorFlow, classification, feature engineering, model evaluation, ROC/AUC, backtesting

LLM Systems: OpenAI, Gemini, Claude, Large Language Models (LLMs), LangChain, CrewAI, prompt design, context filtering, LLM evaluation

Data Science: Pandas, NumPy, R, MATLAB, MNE, ggplot2, statistics, permutation-based FDR, bioinformatics, EEG/fNIRS

Data/Cloud: SQL Server, PostgreSQL, Azure Synapse, Databricks, Delta Lake, AWS, Docker, Power BI, Linux

MACHINE LEARNING, AI, AND DATA EXPERIENCE

CodeAssist – Full-Stack AI Feedback System | GitHub

Aug. 2025 – Present

University of Texas at Austin | Flask, React, PostgreSQL, OpenAI, Gemini, Claude

Austin, TX

- Developed a full-stack AI feedback platform for programming courses, integrating code editing, file uploads, submissions, and AI chat feedback for debugging, algorithm complexity, implementation ideas, and test-case reasoning.
- Designed assignment-level AI controls so instructors can configure prompt templates, permitted context sources, model providers, request limits, and cooldown policies for course-specific feedback.
- Implemented context-filtering pipelines that selectively pass assignment descriptions, code, uploaded files, tests, execution results, and output to OpenAI, Gemini, Claude, or local models.

Enterprise Risk Analyst Intern

May 2025 – Dec. 2025

Toyota Industries Commercial Finance | Python, SQL, Azure, Power BI

Dallas, TX

- Integrated PayNet inquiries and internal loan servicing data through SQL Server ETL, then engineered borrower, contract, and asset features including LTV, tenure, inquiry patterns, vehicle segments, and servicing attributes.
- Trained Logistic Regression and Random Forest credit-risk models, validating ROC/AUC, confusion matrices, and feature importance to achieve **AUC > 0.82** and improve high-risk case identification by **3–4%**.
- Backtested predicted risk buckets against observed outcomes, identifying up to **3.7× higher default rates** in high-risk groups and informing credit scorecard refinements.

Echo – Interactive AI Voice Simulation System

Jun. 2026 – Present

University of Texas at Austin | Mentor: Prof. Bo Xie | Next.js, TypeScript, OpenAI API

Austin, TX

- Built an event-driven voice simulation system that generates context-aware responses from user pauses, errors, skipped fields, and task switching using session state, trigger rules, browser TTS, and OpenAI APIs.
- Designed cooldown, interval, and event-priority logic to prevent repetitive playback and keep generated responses tied to task state and interaction timing.

FinChat – AI-Powered Financial Analysis System

Jan. 2023 – May 2024

Southern Methodist University | Python, FinGPT, LangChain, CrewAI

Dallas, TX (Remote Hybrid)

- Built LLM workflows for financial report analysis, extracting and processing 100+ SEC filings from EDGAR into structured data for metric extraction, trend detection, and company-level comparison.
- Integrated FinGPT, LangChain, and CrewAI components for contextual financial interpretation, improving filing-analysis accuracy by 3% and reducing manual review time.

Data Scientist Assistant Intern – Image Restoration

Jul. 2023 – Sep. 2023

Chinese Academy of Sciences | TensorFlow, Cloud, GFPGAN

Beijing, China

- Deployed image restoration modules on Alibaba Cloud via SSH, implementing VAE loss, Gaussian filters, and LSGAN functions for scalable reconstruction workflows.
- Constructed a TensorFlow 2 CNN, split 3.4k images into train/validation/test sets, and achieved **80% accuracy** across 5 image categories.

RESEARCH PROJECTS AND PUBLICATIONS

Researcher & TA – Biological Data Analytics

Jan. 2024 – Dec. 2024

University of Texas at Austin | R, Linux, AWS, Docker

Austin, TX

- Processed **250GB** of RNA-seq data with Salmon, RSEM, Trinity, and samtools, achieving a **98.7% mapping rate** across reproducible R/Linux pipelines.
- Performed Voom/Limma differential-expression analysis, identified 48 key genes with permutation-based FDR, and automated normalization, PCA, and volcano-plot workflows in R.

Neuroengineering Lab – Machine Learning Research Assistant

Jan. 2023 – May 2024

Python, MATLAB, MNE, RF, SVM, KNN, XGBoost

Remote

- Processed **35GB** of EEG/fNIRS data for signal cleaning, feature extraction, connectivity analysis, statistical testing, and emotion-recognition model evaluation with RF, SVM, KNN, and XGBoost.

Kiwifruit Quality Grading – Machine Learning Member | Paper

Aug. 2022 – May 2024

Huazhong Agricultural University | Computer Vision, Feature Fusion

Wuhan, China (Hybrid)

- Contributed to image-based grading with color, texture, shape, and fusion features; benchmarked feature combinations for a paper published in *Expert Systems with Applications*.

SELECTED PUBLICATIONS

- 5 peer-reviewed publications** across computer vision, neuroscience, and social science research: *Expert Systems with Applications* (2025), *Frontiers in Neurology* (2024), *Frontiers in Neuroscience* (2023), and *IEEE ICIPCA* (2023) on EEG-based emotion recognition using machine learning.